

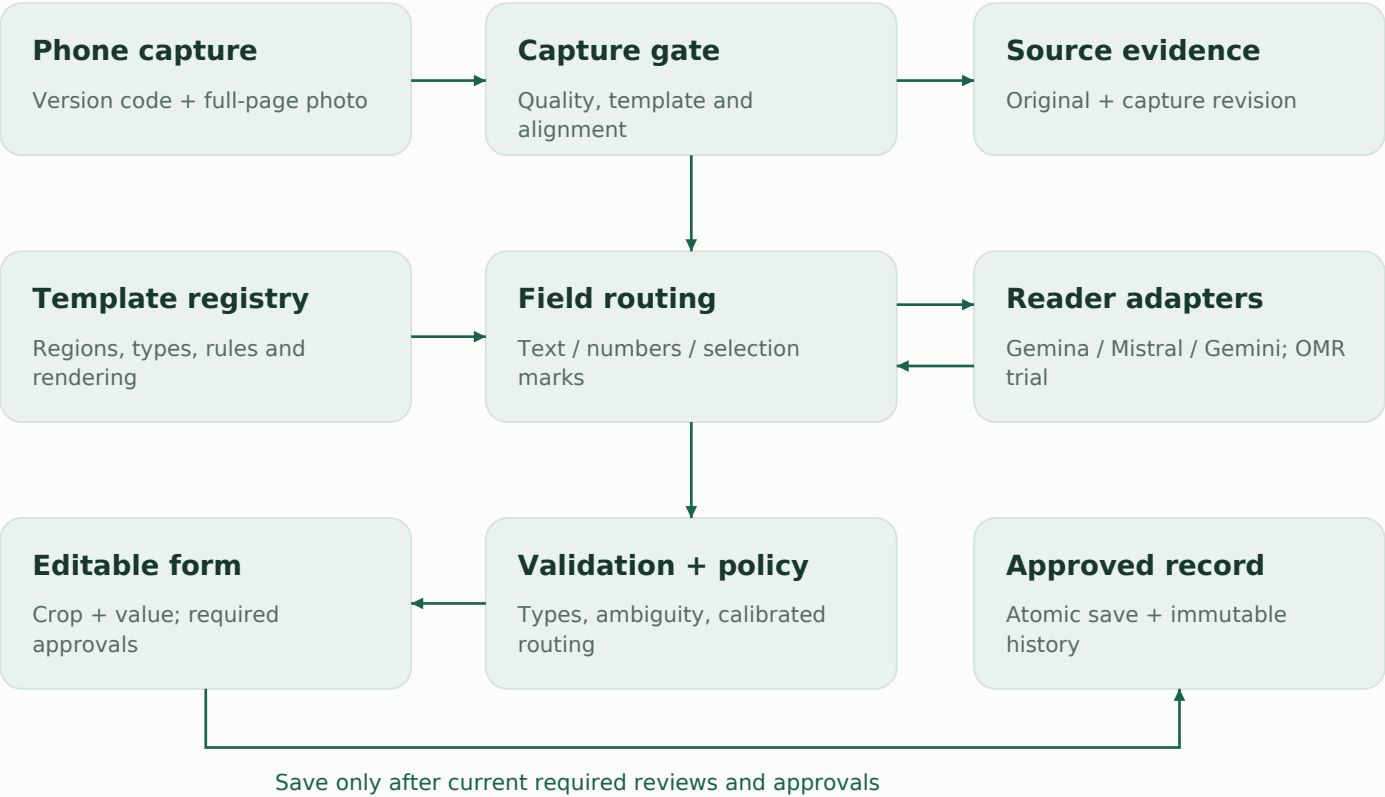
Paper becomes an editable form

Ordinary Hebrew handwriting, numbers and mixed entries. Phone capture, cloud recognition, brief review and explicit confirmation for important fields.

Output. A data-backed Hebrew form that resembles the paper. Fields remain editable; the photograph and field crops provide evidence. The extracted record is the product output.

Revision 2. Add Mistral OCR 4 to the provider trial; compare paper layouts; route fields by type; evaluate missed errors and reviewer effort. These are design decisions, not measured performance claims.

Architecture: shared template, typed extraction, review gate



Scope and remaining uncertainty

Start with one form family and a mobile web application. Preserve normal handwriting; use fixed choices where suitable. A native scanner is an option if phone capture fails the trial. Structured drawings remain a separately evaluated extension.

Still unproven: the best Hebrew recognizer, the most usable digit layout, achievable review time and error rates. No real company forms or live OCR results were supplied. The fence form remains an illustrative example.

A protocol that ordinary writers can use

Use the template to make field meaning explicit. Research on heterogeneous forms supports field types and adequate writing space; the best box layout remains an experiment. [R1]

הזמנת גדר והתקנה

FENCE-01 / v1

דוגמת חכנון

פרטי לקוח

טלפון

050-000-0000

שם הלקוח

יעל לוי

כתובת ההתקנה

דרך הגפן 17

פרטי ההתקנה

סכום כולל - ₪

8450

אורך כולל - מטר

12.5

כולל שער

כן

דגם הגדר

פרטיות לבן

הערות

הערות

להתקשר לפני ההגעה

תאריך: 30.09.2026

סופס לדוגמה בלבד

Paper rules to carry into the pilot

- Identify the form.** A template/version code and four registration marks establish identity and page geometry. Keep the marks clear.
- Give writing room.** Keep printed labels outside entry regions. Use whole-field boxes for names and lined space for sentences.
- Make corrections explicit.** A crossed-out or conflicting entry requires review. Use a labeled correction area or replace the page.
- Capture the full page.** Keep it flat and include the corners. Check clipping, blur, glare and registration before extraction.

Field contract and proposed paper treatment

Field family	Paper + digital rule
Names / addresses	Ordinary Hebrew in a generous box. Preserve literal spelling and mixed numbers; do not silently replace unusual names with dictionary words.
IDs / phone numbers	Trial A: whole-field box. Trial B: one digit per cell. Store strings so leading zeros remain intact. Keep the number order visually unambiguous.
Quantities / dimensions	Print the unit outside the entry. Define one decimal convention and whether separators are allowed. Store raw text, normalized value and unit separately.
Fixed choices	Use explicit checkboxes or bubbles with a documented correction rule. Multiple marks or uncertain marks trigger review; blank is distinct from a negative answer.
Sentences / notes	Use a dedicated lined region. Preserve the transcription and reading order. Track omissions and additions as well as character errors.

TemplateVersion stores page geometry, field polygons, stable field IDs, Hebrew labels, types, units, choices, required/blank rules, normalization rules, review policy and print/screen layout. Each submitted form pins an immutable version.

A/B decision: compare writer effort, extraction accuracy and review time before standardizing digit cells. In the RTL interface, isolate numeric runs so values are displayed in their intended order.

Source evidence beside the editable value

Image-snippet research supports bringing the handwritten field onto the entry screen. Our AI-review workflow must still demonstrate that people catch errors. [R2, R6]

סגירה

אישור חובה

אורך כולל · מטר

שדה חשוב · נדרש אישור גם כשהזיהוי נראה תקין

הכתב המקורי · דוגמה להמחשה

12.5

הערך בטופס הדיגיטלי

12.8

אישור והמשך

- Review policy**
- Ordinary fields.** Show all populated values in the editable form for brief review. Source crops are one tap away.
- Issues needing attention.** Show the source crop with a specific reason: unreadable, ambiguous, missing, invalid, conflicting or low measured reliability.
- Important fields.** Always require explicit confirmation of the current value against the crop, even with a high provider score. No bulk approval for these fields.
- Evidence changes.** Editing a value or replacing its source invalidates the old confirmation. Reruns add candidates; they cannot overwrite a human decision.
- Save gate.** Block final submission until required values and reviews are resolved and each important field has a current confirmation.

Keep these states separate

State	Meaning and allowed action
Readable / valid	A candidate satisfies the field rules. This does not establish that it matches the handwriting.
Blank / unreadable / conflict	Blank: no entry. Unreadable: ink cannot be resolved. Conflict: competing values or corrections. Preserve the state and request the appropriate review.
Reviewed / confirmed	A person resolved an issue or approved a value. Bind that decision to field revision, source revision, template version and reviewer.
Draft / approved record	Incomplete forms can be saved as drafts. Export and final submission require the server gate; an unresolved required value cannot pass.

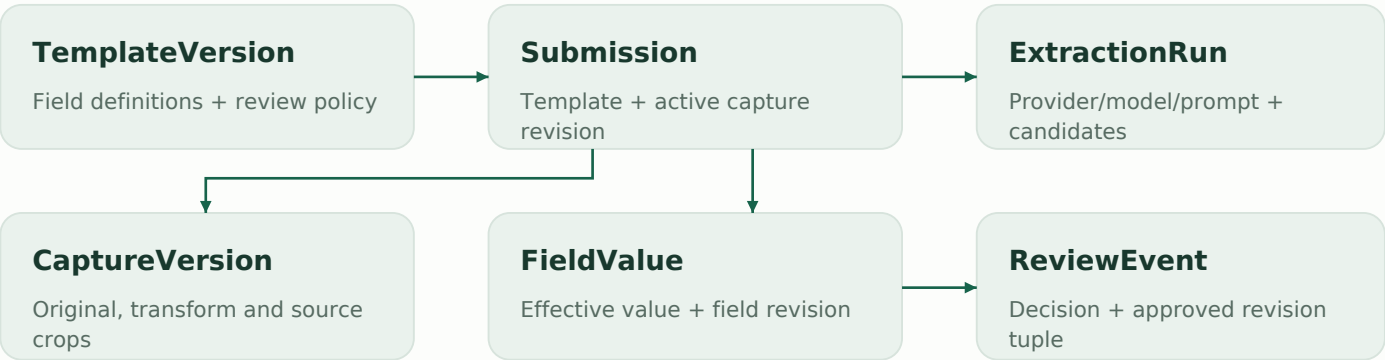
Confidence policy: retain the provider score and its meaning, including whether it applies to a word, block, page or field. Missing confidence stays null. Choose thresholds on calibration data; do not display an unvalidated percentage as a probability of correctness. [T2, T4]

Own the form, evidence and review history

React + TypeScript for the RTL mobile interface; a Python API and worker; PostgreSQL, object storage and a durable queue. Provider integration remains behind a common adapter.

Component	Revised responsibility
Capture + image worker	Preserve the original; identify the template; assess quality; rectify a flat page; save the transform and source-linked crops. Poor geometry returns to capture. [R5, T9]
Template + field router	Supply field labels, types, units and allowed values to extraction. Trial text/number reading separately from optical mark recognition for selections. [R1, T5]
Provider adapters	Support Gemina, Mistral OCR 4 and direct Gemini. Compare full-page extraction with selective crop retries. Normalize candidate records, not provider confidence scales.
Validation + review	Check field rules and cross-field consistency. Enforce explicit important-field approval. Route uncertain values without inventing or silently correcting them.
Persistence + API	Idempotent extraction jobs, immutable candidate runs, revision-checked edits, append-only review events and an atomic final-submit transaction.

Data relationships



Common candidate record

field_id; raw transcription; normalized candidate; blank/readable/unreadable/conflict state; source page and region; capture/template versions; provider/model/run; provider confidence with granularity; validation issues. Keep machine candidates separate from the current human-edited value.

Concurrency: reject stale edits. A confirmation refers to the current value revision and source revision. Retakes invalidate confirmations for affected fields. Retry without duplicating a submission; preserve corrections and review history.

Operations: keep credentials on the server, limit access to source images, configure retention, cap retries and log cost/latency. Roll out provider/model updates only after regression evaluation.

Three candidates, one evaluation contract

Test every candidate against the same field schema, real photographs and verified answers. Documented capability qualifies a service for evaluation; it does not establish accuracy.

Candidate	Why evaluate it	What remains to prove
Gemina [T1]	Documents Hebrew handwriting, RTL, form selections, typed extraction and a review component.	Accuracy on ordinary writers and mixed fields; handling of corrections; useful confidence routing and crop provenance.
Mistral OCR 4 [T2]	Adds a documented alternative with structured annotations, block coordinates and configurable confidence granularity. Hebrew appears in its multilingual evaluation.	The intersection of modern Hebrew handwriting, numeric forms and phone capture has no separately verified result in our research.
Direct Gemini [T3]	Reads images and can return a supplied JSON schema. A baseline for using page context and field descriptions directly.	Literal transcription, omissions, guessed values, reliable abstention and field association. Schema-valid JSON alone is insufficient.

Two stages of comparison

- 1. Controlled baseline.** Use identical input photographs, field definitions and normalization rules. Record the exact model version, prompt/schema version and settings. Allow only documented, bounded retries; count every failure and abstention.
- 2. Practical pipeline.** On development data, compare full-page context against context plus selective field crops. Trial OMR for fixed choices. Then freeze each complete configuration and score the held-out forms. Compare cost and latency as well as correctness.

Selection and fallback rules

Choose the simplest configuration that meets the predeclared field-risk and review-time gates. A second reading is an experiment for doubtful fields, not a default ensemble. Agreement between readers is not proof of correctness; both may make the same mistake.

If a field family remains unreliable, change its paper protocol or require direct transcription for that field. If necessary, evaluate a trainable system such as Kraken using suitable Hebrew data. Keep human review available when services fail. [T6]

Shortlist boundaries

Azure Read/Layout and Google Cloud Vision list Hebrew for general/printed OCR, but their published handwriting tables omit Hebrew. Do not select those products on a generic language-support claim. This finding does not describe Gemini. [T7, T8]

Measure usable fields and remaining errors

The 30-form batch is a feasibility screen. Public Hebrew datasets supplement it; they cannot establish the reliability of the complete phone-to-form workflow. [R3, R4]

Dataset and split rules

Feasibility batch. Start with 30 completed forms from at least 6 writers on representative phones. Include Hebrew names, mixed addresses, IDs, decimals, notes, blanks, corrections and deliberately difficult capture conditions.

Paper A/B test. Compare whole-field number boxes with digit cells on balanced form variants. Counterbalance variant order across writers. Record writing time, corrections and ease of use as well as extraction and review time.

Separate tuning from evidence. Split by writer. Keep all crops, photos and layout variants of the same physical form in one split. Use development data for prompts, crop strategy and confidence thresholds; lock the final test set before evaluating.

Gold answers. Two people transcribe fields independently of model suggestions; adjudicate disagreements. Label genuinely unreadable entries explicitly. Store field type, source region and correction state. Add fresh writers and conditions for a larger acceptance set.

Measure	Definition / decision use
Exact field match	Correct complete values / all applicable fields, reported by type and risk. Predeclare harmless normalization. Keep wrong, blank, unreadable and failed outcomes visible.
Review routing	Error escape rate: wrong predictions not flagged / all wrong predictions. Also report wrong unflagged / all unflagged fields, and the share of fields sent to review.
Post-review errors	Wrong final values / reviewed values, separately for important fields. Measure against adjudicated truth; a confirmation click is not correctness evidence.
Human effort	Writing time; median and slow-case review time; manual edits; retakes; abandonment. Compare AI-assisted review with source-crop manual entry on balanced cases.
Pipeline performance	End-to-end latency, capture rejection/retake rates, retries and cost per approved form including human time. Count full extraction failures, not only returned fields.

Acceptance: agree field-specific error limits, review-time budget, fallback and cost cap before the locked test. Report sample counts and uncertainty. A small batch cannot validate rare errors: expand evidence for important fields before rollout, and keep explicit confirmation mandatory. [R1, R2, T4]

Evidence gates determine the next build

Test recognition, paper usability and review effectiveness before full production integration. Each phase has a concrete deliverable and a decision gate.

Phase	Concrete deliverables	Exit condition
A. Contract	Field inventory; importance policy; blank/correction rules; template versions A/B; normalized record schema; proposed business targets.	Every field has an agreed meaning, type, unit, review policy and output location.
B. Test material	30-form feasibility batch; writer metadata; capture variants; independent gold answers; fixed development/test split.	Coverage includes the difficult field types and real capture conditions. No writer/form leakage between tuning and test.
C. Reader trial	Three provider adapters; one evaluation harness; baseline and crop-strategy results; paper A/B and review-time comparison.	A frozen configuration is promising enough for a pilot. If not, revise protocol or use manual entry for weak fields.
D. Working MVP	Capture gate; editable RTL replica; field evidence; issue resolution; important-field approval; drafts; job retries; atomic submit.	Required unresolved fields cannot submit. Edits and retakes invalidate approvals; retries and stale clients preserve human work.
E. Acceptance pilot	Larger fresh-writer test set; predeclared targets; error audit; monitored field rollout; documented manual fallback.	Meet field-specific risk and effort targets with adequate evidence. No high-confidence shortcut around critical approval.
F. Drawing extension	A fixed symbol grammar; labeled graph examples; topology editor; separate topology/dimension evaluation.	Correct objects, connections and dimension links are verified. Text accuracy alone cannot pass the drawing gate.

Engineering checks that matter

Exercise the final-submit gate with missing required values, unresolved conflicts and stale confirmations. Verify that an edit, retake or template mismatch cannot reuse an old approval. Simulate failed jobs and duplicate submissions. Check mixed Hebrew/numeric rendering on the actual target phones.

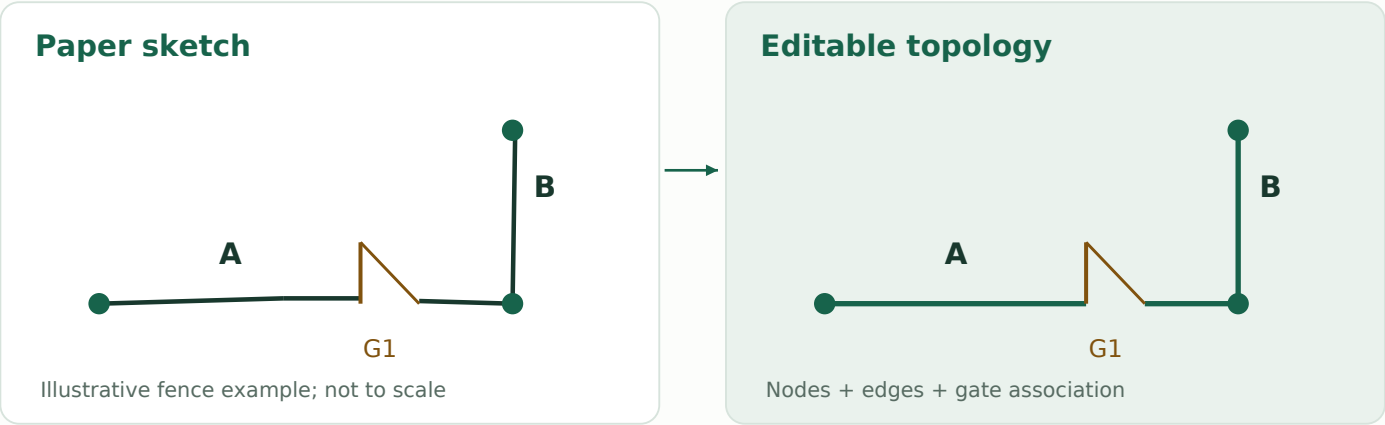
Immediate handoff

Collect one blank form and representative completed examples. Mark the always-confirm fields. Define allowed number formats and drawing scope. Then produce the two paper variants, gold-answer sheet and provider-comparison harness.

Decision log: record why a layout, reader or threshold was selected. A change to model, prompt, schema or capture processing triggers a relevant regression check before rollout.

Recognize objects and their connections

Diagram-recognition research supports separate symbol and relationship recognition. Define our drawing grammar and evaluate its topology independently of text. [R7]



Paper protocol	Structured meaning / validation
Labels A, B, C for segments	Each segment is an edge with endpoint references. Match the label to a row in the measurements table.
Dots at connections	Explicit junction nodes. Crossings without dots are not automatically connected; uncertain connectivity requires review.
Defined gate symbol + G1	A gate attached to a named segment, with an opening width and any required swing direction.
Separate measurement table	Length, unit and segment label. Explicitly define whether a dimension includes the gate opening.
Source-linked graph editor	Review unmatched labels, missing/extra objects, disconnected endpoints, disputed junctions and dimensions linked to the wrong segment.

Data model and acceptance

Node: coordinates and type. **Segment:** endpoints, label and measurement link. **Gate:** segment, position and opening. **Measurement:** raw text, typed value, unit, scope, source and review state. Preserve source evidence for each recognized object and relation.

Score object detection, correct endpoint connections, dimension-to-segment association and whole-sketch correctness separately. Test intersections, gaps, crossed-out labels and ambiguous gate placement. A useful trace of the ink can still encode the wrong structure.

Written dimensions govern physical lengths. Pixel distances provide relative placement unless an explicit, validated scale protocol is introduced.

What the research supports

Sources checked through Tavily MCP on 30 September 2026. Evidence supports design choices; none of these sources establishes production accuracy for our complete workflow.

R1. Field typing for improved recognition on heterogeneous handwritten forms (2019)

Real French accident forms; field types, synthetic data and augmentation improve recognition. Supports typed fields and whole-field metrics; does not establish Hebrew performance.

[Paper](#) | [Implementation](#)

R2. Improving Form-Based Data Entry with Image Snippets (2013)

A controlled study with 26 participants found faster manual entry using snippets without a detectable accuracy loss. Supports crop/value proximity; AI-review effectiveness still needs testing.

[Original paper](#)

R3. The HHD Dataset (2020)

Approximately 1,000 handwritten Hebrew forms scanned at 600 dpi. The linked public HHD_v0 release is isolated characters, not a complete phone-form benchmark.

[Paper](#) | [Public v0 release](#)

R4. ivrit.ai Hebrew handwriting benchmark (2026)

225 verified lines across 10 pages, one page per writer; intended as a held-out test set. Relevant to modern Hebrew but too small and narrow to establish our form workflow.

[Dataset card](#)

R5. SmartDoc-QA (2015)

Phone-captured documents under controlled lighting, blur and perspective distortions. Supports capture-condition testing; its older OCR scores are not current performance estimates.

[Original paper](#)

R6. Integrating ODK Scan into the CHW supply chain in Mozambique (2013)

Field deployment of template alignment, automatic mark reading and snippet-assisted transcription. Handwritten text relies on a person in this system.

[Original paper](#) | [Form design guide](#)

R7. Arrow R-CNN for handwritten diagram recognition (2021)

Joint recognition of diagram symbols and structure. Supports topology as a separate target; the evaluated diagram families are not our future installation sketches.

[Published paper](#) | [Associated datasets](#)

Tools, guides and documented limits

Vendor documentation establishes interfaces and claimed capabilities. Provider choice, review thresholds and paper protocol remain subject to the experiments in this plan.

T1. Gemina

Hebrew handwriting/forms and a supplied verification UI. Keep editable records and revision-aware approvals in our application.

[Forms](#) | [API and verification](#)

T2. Mistral OCR 4

Structured annotations, block regions and configurable confidence granularity. General Hebrew evaluation is not a handwriting-on-forms benchmark.

[Release](#) | [Annotations](#) | [API](#) | [Cookbook](#)

T3. Gemini API

Image inputs and JSON schema outputs. Validate extracted meaning and literal values separately from JSON structure.

[Image input](#) | [Structured output](#)

T4. Google Document AI evaluation

Precision, recall and confidence-threshold methodology. Use it as an evaluation reference, independently of provider choice.

[Evaluation guide](#)

T5. OMRChecker

Custom templates, mark recognition and phone alignment markers. Suitable reference for selections; not a Hebrew text reader.

[Project](#) | [Template guide](#)

T6. Kraken

Trainable segmentation/recognition and RTL/BiDi support. A customization route that still needs suitable Hebrew data and validation.

[Overview](#) | [Training tutorial](#)

T7-T8. Google Cloud Vision and Azure Read/Layout

Both document general Hebrew OCR; their specific handwriting support tables omit Hebrew. Avoid extrapolating from language counts.

[Google languages](#) | [Azure languages](#)

T9-T10. OpenCV and ML Kit capture

Homography aligns a flat page; it does not solve arbitrary page curvature. A native capture adapter is conditional on the phone trial.

[OpenCV homography](#) | [ML Kit scanner](#)